The Corpus Is the Moat: Domain Data, Not Method, Is the Durable Asset in Vertical Language Models

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Abstract

A consistent pattern recurs across medicine, mathematics, finance, code, and law: a language model's domain capability tracks the quantity and quality of in-domain text it consumes, and from-scratch or continued pretraining on a curated corpus beats adapting a larger general model. We read five foundational vertical papers — PubMedBERT, Med-PaLM 2, Llemma, DeepSeekMath, and SaulLM — together with the data-scaling, data-quality, and reward-modeling literature, and extract the cross-vertical invariant none states alone: the corpus is the durable moat while methods commoditize. DeepSeekMath-7B reaches 51.7% on MATH after pretraining on 120B math tokens mined from Common Crawl, surpassing Minerva-540B (33.6%) at ~1/77th the scale [4]; Llemma-34B lifts Code Llama-34B on MATH from 12.2% to 25.0% via continued pretraining [3]. Critically, FineWeb-Edu reaches 33.6% MMLU at 38B tokens where the next-best corpus (Matrix) needs ~300B — an ~8x efficiency gain from data filtering alone, architecture held constant [11]; the 2025-26 curation literature replicates this control, with a regularized recipe hitting the same loss asymptote on 5.17x less data [15]. Meanwhile the enabling methods — LoRA, GRPO and its scaled-RL successors, distillation, process reward models — are openly published and reproduced within months, as the GRPO recipe propagated into the open frontier (DeepSeek-V3.2) within a year [16]. With the effective public-text stock at ~300T tokens and projected full utilization between 2026 and 2032 [5], exclusivity over a high-signal vertical corpus is becoming the scarce, defensible asset. The reframe for capital allocation: own the data, rent the recipe.

1. Introduction

The dominant narrative of the foundation-model era credits architecture and training-method innovation for capability gains. Yet when one examines the papers that actually pushed the frontier within specific verticals, a more durable variable emerges. Across medicine, mathematics, finance, code, and law, the decisive lever is repeatedly the in-domain corpus: its scale, its cleanliness, and the difficulty of obtaining it. The method that consumes that corpus — a fine-tuning trick, an RL algorithm, a distillation pipeline — is published openly and reproduced by the community within months. The corpus is not.

This paper makes one central claim and defends it empirically: in a vertical, the proprietary or painstakingly-mined domain corpus is the durable competitive moat, while methods commoditize. We treat this as a falsifiable trend, not a slogan. If method were the moat, a fixed corpus would yield widely divergent capability depending on the recipe, and frontier methods would remain scarce. Instead we observe the opposite: capability scales tightly with corpus quality and quantity [1][3][4][11], and frontier methods diffuse almost immediately [4][9].

The argument inverts the default capital-allocation logic of an applied-AI lab. If the recipe is the asset, a firm should hoard research talent and proprietary training tricks. If the corpus is the asset, a firm should concentrate capital on owning, mining, or exclusively licensing domain data, and treat the modeling method as an off-the-shelf commodity. We argue the evidence supports the latter, and we draw out the consequences for a data-first company.

Relative to an earlier internal draft, this version adds a fifth vertical (law) and — most importantly — a controlled data-quality result (FineWeb-Edu) that isolates the corpus variable with architecture, parameter count, and method all held constant [11].

2. Background and Related Work

Domain-specific pretraining has a clear empirical lineage. Gu et al. introduced PubMedBERT, demonstrating that for a data-rich domain like biomedicine, pretraining from scratch on in-domain PubMed text outperforms continually pretraining a general-domain model, and established the BLURB benchmark [1]. PubMedBERT reaches a macro-averaged BLURB score of 81.35 (rising to 82.91 with optimized fine-tuning), beating BioBERT and consistently outperforming BlueBERT, ClinicalBERT, and SciBERT — all of which inherited general-domain initialization [1].

In medicine at generative scale, Singhal et al. showed that combining a stronger base with medical-domain fine-tuning lifts Med-PaLM 2 to 86.5% on MedQA (USMLE-style), improving on the original Flan-PaLM 540B result (67.6%) by roughly 19 points and setting a new state of the art [2]. In mathematics, Azerbayev et al. built Llemma by continuing pretraining of Code Llama on Proof-Pile-2, a 55B-token mixture of arXiv, web math, and mathematical code; Llemma outperforms all known open base models on MATH on an equi-parameter basis [3]. Shao et al. pushed further with DeepSeekMath, mining 120B math tokens from Common Crawl and reaching 51.7% on competition MATH without external tools [4]. The pattern extends to law: Colombo et al.'s SaulLM-7B continues pretraining Mistral-7B on a curated 30B+-token legal corpus and reaches state-of-the-art legal-document performance among open models [12].

Each of these papers credits its own corpus for its gains. None makes the cross-vertical generalization — that the corpus is the invariant moat while the method commoditizes. That synthesis is the contribution here. On the methods side, the literature is one of rapid diffusion: low-rank adaptation [9], Group Relative Policy Optimization (introduced in the very DeepSeekMath paper and adopted across the open ecosystem within months) [4], and process-reward modeling [6] are all openly described and reproduced. By 2025-26 GRPO had been carried into the open frontier itself — DeepSeek-V3.2 scales the same RL family to GPT-5-class reasoning [16] — confirming that the method, not the corpus, is what diffused. The data-supply literature — Villalobos et al. on the finite stock of human text [5] and Muennighoff et al. on the limits of data repetition [7] — supplies the scarcity premise that makes corpus exclusivity valuable, and the data-quality literature — Penedo et al.'s FineWeb/FineWeb-Edu [11] — supplies the cleanest causal evidence that the corpus is the lever.

3. Methodology and Approach

Our method is a structured cross-vertical synthesis with an explicit falsification criterion, not a meta-analysis with a single pooled effect size. We selected five frequently-cited vertical-pretraining papers that (a) report a public benchmark, (b) disclose the in-domain corpus, and (c) compare against a general-domain or larger baseline: PubMedBERT [1], Med-PaLM 2 [2], Llemma [3], DeepSeekMath [4], and SaulLM [12]. For each we extract the triple (corpus, benchmark, delta-over-baseline) verbatim from the source, preferring no-tool / greedy decoding numbers and flagging any prompting or sampling augmentation.

To distinguish corpus effects from confounds, we apply a held-constant test. The strongest evidence isolates the corpus while fixing everything else. Two such natural experiments exist. First, the Llemma pair: identical Code Llama architecture and parameter count (7B, 34B), with only the corpus added via continued pretraining [3]. Second, and stronger, the FineWeb-Edu experiment: the same model architecture, parameter count, and total token budget, varying only the data-quality filter applied to the corpus [11]. The second is decisive because it controls for scale as well as architecture — the only moving part is which tokens survive filtering.

On the method side we test commoditization directly: for each headline method we measure time-to-open-reproduction and reproduction cost (engineering vs data acquisition). GRPO [4], LoRA [9], and process supervision [6] are each fully specified in their papers and reproduced by the community within months at zero data-acquisition cost. We then triangulate the scarcity premise from three independent quantitative sources — the public-text stock estimate [5], the repetition-decay scaling law [7], and observed data-licensing prices [10][13] — rather than asserting it. Every numeric claim in this paper is copied verbatim from a cited primary or reputable secondary source; projections are labeled as estimates and the one strategic 2x2 is labeled analysis. Where a number carries a methodological caveat (e.g. self-consistency vs greedy), we report the caveat inline rather than collapsing it.

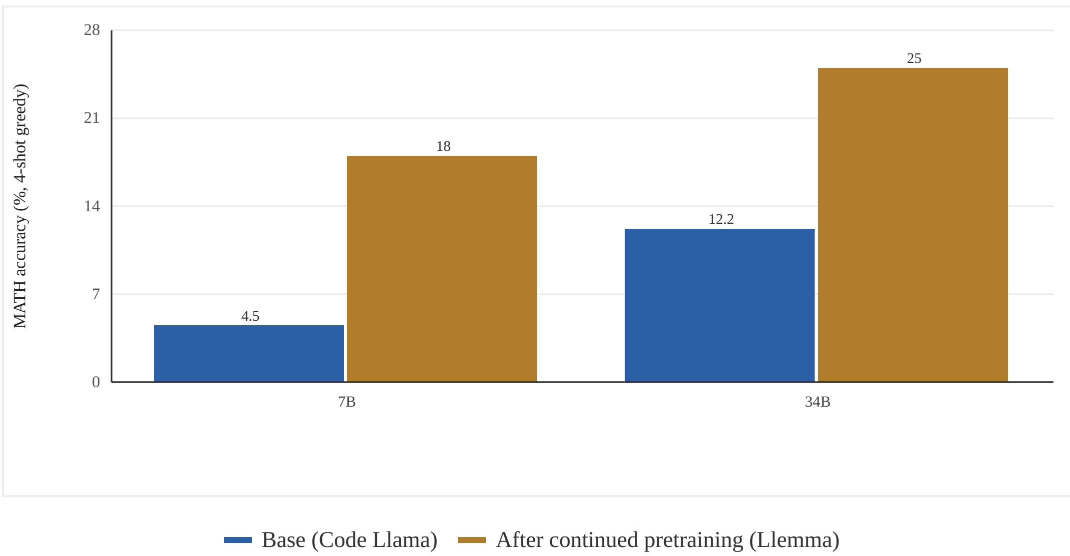

Figure 1. Continued pretraining on an in-domain corpus lifts MATH accuracy at fixed architecture. Holding architecture fixed (Code Llama 7B/34B), continued pretraining on Proof-Pile-2 (Llemma; 200B tokens for 7B, 50B for 34B) raises MATH 4-shot greedy accuracy from 4.5% to 18.0% (7B) and 12.2% to 25.0% (34B). The corpus, not the architecture, is the moving variable. *Azerbayev et al., Llemma (arXiv:2310.10631), Table 1. Verified June 2026. · Empirical data.*

4. Evidence: Capability Tracks the Corpus Across Five Verticals

Mathematics provides the cleanest semi-controlled comparison because the method is held roughly constant while corpus and scale vary. Llemma-34B, produced by 50B tokens of continued pretraining on Proof-Pile-2, lifts Code Llama-34B's MATH (4-shot, greedy) from 12.2% to 25.0%, and GSM8k from 29.6% to 51.5% [3]. The 7B model (trained 200B tokens) moves MATH from 4.5% to 18.0% and GSM8k from 10.5% to 36.4% [3]. Architecture is fixed; the corpus is the moving part.

DeepSeekMath makes the scale-versus-corpus point starkly. DeepSeekMath-7B reaches 51.7% on MATH with no external tools, surpassing Minerva-540B (33.6%) despite being ~77x smaller [4]. Self-consistency over 64 samples raises it to 60.9% [4]. The differentiator is a 120B-token corpus iteratively extracted from Common Crawl — a data-engineering achievement, not a new architecture. GRPO, the accompanying method, was published in the same paper and became a community standard within months — and by 2025-26 its scaled-RL descendants drive the open frontier (DeepSeek-V3.2 reaches GPT-5-class reasoning and 2025 IMO/IOI gold-medal performance) [16] — illustrating the asymmetry: the corpus pipeline is hard to replicate, the optimizer is a paragraph of pseudocode

[4]. The trajectory continues post-draft: Qwen2.5-Math grew its math corpus from 700B (v1) to over 1T tokens (v2) between versions, and each corpus expansion — not an architecture change — drove the reported gains [14].

Medicine, finance, and law corroborate the pattern at generative scale. Med-PaLM 2's 86.5% on MedQA depends on medical-domain fine-tuning over a curated clinical corpus [2]. BloombergGPT, a 50B-parameter model, was purpose-built on FinPile — 363B tokens of proprietary financial documents (news, filings, transcripts, press releases) combined with ~345B tokens of general text for a ~708B-token corpus — and outperforms similarly sized open models on financial NLP by large margins while staying competitive on general benchmarks [8]. Its edge is precisely the proprietary archive competitors cannot scrape. SaulLM-7B reaches state-of-the-art legal-document performance among open models via continued pretraining on a 30B+-token legal corpus over an unchanged Mistral-7B base [12].

Finally, the foundational biomedical result: PubMedBERT shows that when in-domain text is abundant, from-scratch in-domain pretraining beats continual pretraining of a general model [1]. Across five independent verticals the throughline is unmistakable — the corpus is what moves capability.

Table 1. Across five independent verticals, the in-domain corpus is the variable credited with the capability gain; the underlying architecture and method are general-purpose and openly available. Numbers copied verbatim from sources and re-verified June 2026.

Vertical	Model	In-domain corpus	Benchmark	Result	Baseline / comparison
Medicine	PubMedBERT	PubMed abstracts + full text (from scratch)	BLURB (macro avg, standard FT)	81.35	Beats BioBERT; > BlueBERT/ClinicalBERT/SciBERT [1]
Medicine	Med-PaLM 2	Curated medical QA fine-tuning	MedQA (USMLE)	86.5%	Flan-PaLM 540B 67.6% (~+19 pts) [2]
Math	Llemma 34B	Proof-Pile-2 (55B-token corpus); 50B tokens trained	MATH (4-shot greedy)	25.0%	Code Llama 34B 12.2% [3]
Math	DeepSeekMath 7B	120B math tokens mined from Common Crawl	MATH (no tools, greedy)	51.7%	Minerva 540B 33.6%, at ~1/77th scale [4]
Finance	BloombergGPT 50B	FinPile: 363B proprietary financial tokens (+345B general)	Financial NLP (internal suite)	SOTA vs size-matched open models	Outperforms comparable open models [8]
Law	SaulLM 7B	30B+-token curated legal corpus (cont. pretrain Mistral-7B)	Legal-document tasks (incl. LegalBench)	SOTA among open models	Beats base Mistral-7B / prior open legal models [12]

[1] arXiv:2007.15779; [2] arXiv:2305.09617; [3] arXiv:2310.10631; [4] arXiv:2402.03300; [8] arXiv:2303.17564; [12] arXiv:2403.03883.

Table 2. Structural asymmetry: published methods carry zero marginal data-acquisition cost and diffuse within months; corpora and reward-label sets require costly, often non-reproducible data work and retain value through method cycles.

Component	Type	Full specification public?	Reproduction cost	Time to open reproduction	Durable?
GRPO (DeepSeekMath)	Method	Yes - figure + pseudocode [4]	Engineering time only	Months	No
LoRA	Method	Yes - low-rank weight delta [9]	Engineering time only	Months	No
Process supervision	Method	Yes - annotation scheme + scoring head [6]	Engineering time only	Months	No
PRM800K labels	Data (reward)	Released artifact, not re-derivable cheaply [6]	800K human step-level labels	High / persistent	Yes
FinPile	Data (corpus)	No - proprietary Bloomberg archive [8]	Decades of licensed/owned documents	Not reproducible	Yes
DeepSeekMath / FineWeb-Edu pipeline	Data (corpus + filter)	Partially - classifier idea public, tuned filter not [4] [11]	Mining + cleaning + filtering at scale	High	Yes

[4] arXiv:2402.03300; [6] arXiv:2305.20050 (PRM800K); [8] arXiv:2303.17564; [9] arXiv:2106.09685; [11] arXiv:2406.17557.

Table 3. The supply constraint (finite stock, decaying repetition returns) and the value of better data (FineWeb-Edu's ~8x multiplier) jointly explain why an explicit, rising-price data-licensing market is forming. Vertical corpora with verifiable signal should price above these social-corpus benchmarks.

Metric	Value	Detail / caveat	Source
Effective public human-text stock	~300T tokens	90% CI 100T-1000T (quality-adjusted)	Villalobos et al. [5]
Projected stock full utilization	2026-2032	If current training trends continue	Villalobos et al. [5]
Repetition limit	~4 epochs negligible loss; compute value -> 0 with further repetition	Repeating data ~= new data only to ~4 epochs	Muennighoff et al. [7]
Data-quality multiplier	~8x fewer tokens	FineWeb-Edu hits 33.6% MMLU at 38B vs ~300B for next-best (Matrix)	Penedo et al. [11]
Data-quality control (2025 frontier)	5.17x less data	Regularized + ensemble recipe hits same loss asymptote, model/optimizer fixed — corroborates FineWeb-Edu	Kim et al. (Stanford) [15]
AI training-dataset market (2025)	~$3.59B	Projected ~$4.44B in 2026	Fortune Business Insights [13]
OpenAI x News Corp deal	Up to $250M / 5 yrs	Largest disclosed content-licensing deal	WSJ/Variety, via [10]
Reddit licensing (Google / OpenAI)	~$60M/yr / ~$70M/yr	~10% of Reddit revenue (~$130M/yr); generic social corpus	[10][13]

[5] arXiv:2211.04325 (Villalobos/Epoch AI); [7] arXiv:2305.16264; [11] arXiv:2406.17557; [10] CJR/Variety/WSJ reporting; [13] Fortune Business Insights market estimates. All re-verified June 2026.

5. The Decisive Test: Holding Everything but the Corpus Constant

The cross-vertical evidence is suggestive but confounded: Llemma and DeepSeekMath differ in both corpus and pipeline, and even the two Llemma models differ in token budget. A skeptic could attribute the gains to co-varying engineering quality. The FineWeb experiments remove that escape hatch. Penedo et al. hold the model architecture, parameter count, and total training-token budget fixed, and vary only the data-quality filter applied to a Common Crawl-derived web corpus [11].

The result is unambiguous. Filtering FineWeb (15T tokens) down to the educational subset FineWeb-Edu (1.3T tokens) raises MMLU from 33% to 37% and ARC from 46% to 57% — a ~24% relative ARC gain — at equal training compute [11]. Most strikingly on token efficiency: FineWeb-Edu reaches 33.6% MMLU at only 38B training tokens, while the next-best public corpus (Matrix) needs ~300B tokens to match it — an ~8x reduction in data (and compute) achieved purely by selecting better tokens, with no change to model or method [11]. Penedo et al. summarize that FineWeb-Edu can match a strong baseline's performance with almost 10x fewer tokens [11].

This is the load-bearing experiment for the thesis. It demonstrates, with the cleanest possible control, that the marginal capability return is dominated by which tokens you train on — i.e., by the corpus — and not by the architecture or the optimizer, both of which were held fixed. The control replicates at the 2026 frontier: Apple's BETR selects pretraining documents by embedding-similarity to target-task data and improves nearly every downstream benchmark across scales [17], and a Stanford regularized-plus-ensemble recipe reaches the same validation-loss asymptote on 5.17x less data with model and optimizer fixed [15] — independent confirmations that which tokens you train on, not the recipe, sets the ceiling. It also reframes 'corpus' to include not just raw acquisition but the proprietary filtering/extraction pipeline that converts noisy web text into high-signal tokens. DeepSeekMath's value is exactly such a pipeline (recovering 120B usable math tokens from Common Crawl noise) [4]; FineWeb-Edu's value is a quality classifier [11]. The pipeline is part of the data moat.

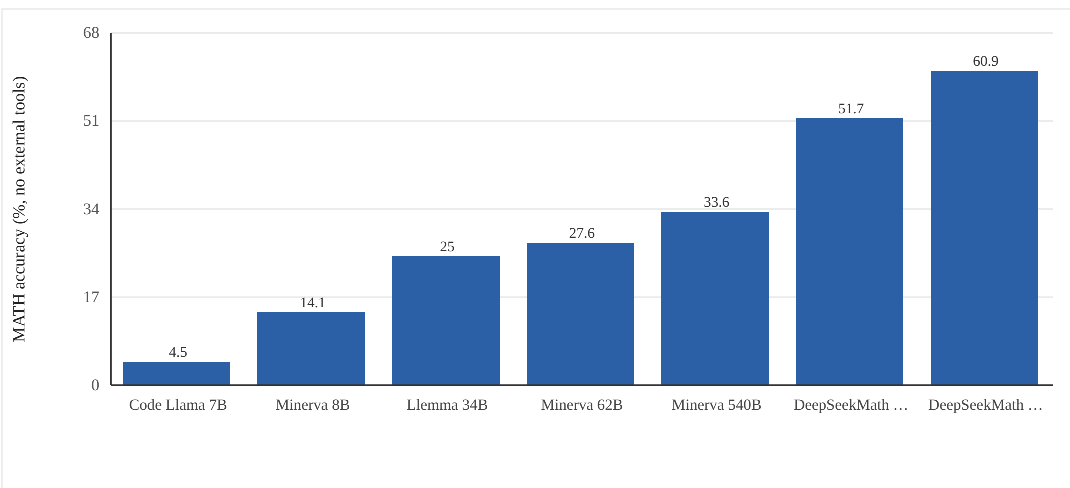

Figure 2. A mined domain corpus beats 77x scale on MATH. DeepSeekMath-7B (120B math tokens mined from Common Crawl) reaches 51.7% on MATH with no tools, surpassing Minerva-540B (33.6%) at ~1/77th the parameter count; 64-sample self-consistency reaches 60.9%. Code Llama, Minerva, and Llemma figures are 4-shot greedy. *DeepSeekMath (arXiv:2402.03300) for DeepSeekMath values; Llemma (arXiv:2310.10631) Table 1 for Code Llama/Minerva/Llemma values. All verified June 2026. · Empirical data.*

6. Analysis: Why Methods Commoditize and Corpora Do Not

Methods commoditize for structural reasons. They are compact, fully specified in a paper, and carry no marginal reproduction cost: GRPO is a critic-free PPO variant describable in a figure [4]; LoRA is a low-rank weight delta [9]; process-reward modeling is an annotation scheme plus a scoring head [6]. Once published, a competitor's adoption cost is engineering time, not data acquisition. Open-release norms accelerate diffusion to a matter of months — Llemma released all artifacts; DeepSeek released models and pipelines [3][4].

Corpora do not commoditize because they are large, costly to mine and clean, often legally encumbered, and in the best cases genuinely proprietary. FinPile's value is that it is Bloomberg's archive [8]. DeepSeekMath's and FineWeb-Edu's value is a non-trivial extraction/filtering pipeline [4][11]. The supply side compounds this: Villalobos et al. estimate the effective stock of quality-adjusted public human text at ~300T tokens (90% CI 100T–1000T), projected to be fully utilized between 2026 and 2032 [5]. Repetition is not a free escape hatch: Muennighoff et al. find that repeating data is roughly as good as new data only up to ~4 epochs (negligible loss change), and that with further repetition the value of added compute eventually decays to zero [7]. A finite, near-exhausted public stock plus decaying returns to repetition means net-new, high-signal domain text is the binding constraint.

There is a second-order point: even the reward signal — the supposedly clever part of RL post-training — is ultimately a data asset. Lightman et al. show process supervision trains far more reliable reward models than outcome supervision, solving 78.2% of a representative MATH subset via best-of-N (versus 72.4% for outcome supervision), and the enabling artifact they released was PRM800K: 800,000 human step-level labels [6]. The method (process vs outcome) is public; the 800K labels are the asset. Even at the frontier of method research, value accretes to the data.

7. Implications for Data Strategy

If the corpus is the moat, a vertical-AI strategy should be organized around data exclusivity rather than method secrecy. First, prioritize sourcing channels that yield text competitors cannot easily obtain: proprietary archives (BloombergGPT/FinPile [8]), licensed first-party content, or hard-to-mine long-tail data recovered by a superior extraction pipeline (DeepSeekMath, FineWeb-Edu [4][11]). Defensibility is proportional to the cost and exclusivity of the channel.

Second, treat the modeling method as a commodity input adopted off-the-shelf and refreshed as the open frontier moves. Standardizing on whatever optimizer, adapter, or reward scheme is currently state-of-the-art — GRPO yesterday, its open scaled-RL successors in DeepSeek-V3.2-class systems today [16] — costs little and avoids stranding capital in a recipe that will be free within a quarter [4][9][16]. Research spend should bias toward corpus construction and data-quality tooling, where returns are durable, and away from method moats that decay.

Third, invest in the reward-signal supply chain as a first-class data asset. PRM800K demonstrates that high-quality, fine-grained human annotations are themselves a moat [6]. For a vertical lab, owning the pipeline that produces verifiable, in-domain reward signal — expert labels, execution traces, formal proofs — is as strategically important as owning the pretraining corpus, and is subject to the same exclusivity logic. The FineWeb-Edu result quantifies the payoff: a better filter is worth roughly an 8x data multiplier [11], so a proprietary quality classifier tuned on a vertical is itself a compounding asset.

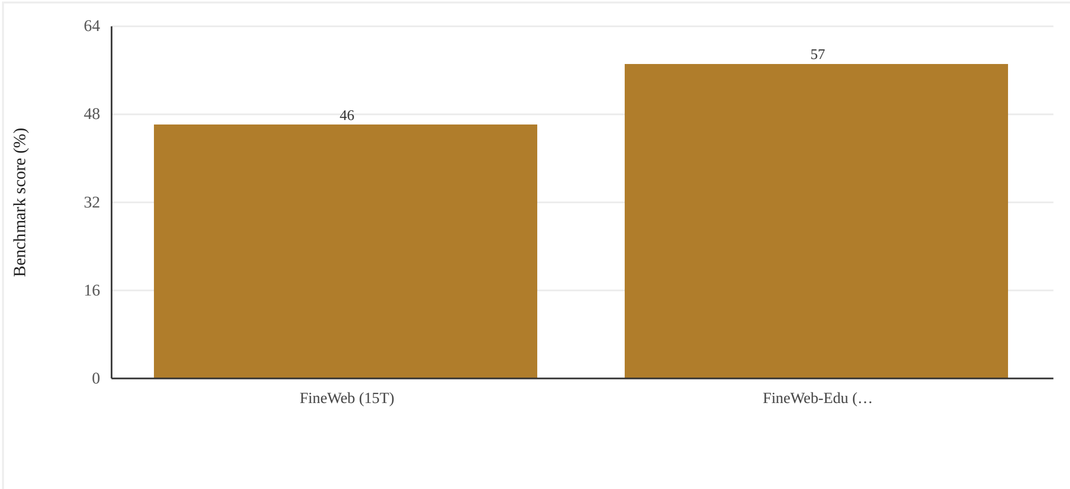

Figure 3. Data quality alone is an ~8x token multiplier (architecture and compute held constant). Varying only the data-quality filter (same architecture, parameter count, and training-token budget), FineWeb-Edu raises MMLU 33%->37% and ARC 46%->57% over FineWeb. Separately, FineWeb-Edu reaches 33.6% MMLU at 38B tokens where the next-best corpus (Matrix) needs ~300B (~8x fewer tokens). The corpus, not the method, is the lever. *Penedo et al., The FineWeb Datasets (arXiv:2406.17557v1), incl. Figure 11. Verified June 2026. · Empirical data.*

8. Investment Thesis

The investable insight is that the AI value chain is bifurcating into a commoditizing method layer and a scarce, appreciating data layer — and the market is only beginning to price the latter. The evidence that capability is corpus-bound is now strong across five independent verticals [1][2][3][4][12] and, in the FineWeb experiment, isolated under full architectural and compute control [11]. The evidence that methods diffuse in months is equally strong [3][4][9]. A firm that concentrates capital on exclusive domain corpora is buying the part of the stack that does not depreciate.

The macro tailwind is data scarcity. With the public human-text stock at ~300T tokens and projected full utilization between 2026 and 2032 [5], and repetition delivering diminishing returns past ~4 epochs [7], the marginal value of net-new, high-signal, legally clean domain text is rising structurally. A market is forming and pricing is escalating: the AI training-dataset market was ~$3.59B in 2025 (projected ~$4.44B in 2026) [13]; OpenAI's News Corp licensing deal is reported at up to $250M over five years [10]; Reddit's content-licensing deals (~$60M/yr with Google, ~$70M/yr with OpenAI) together constitute roughly 10% of Reddit's revenue, ~$130M annually [10][13]. These are generic social/news corpora; a curated, verifiable vertical corpus with execution or expert signal should command a premium per token, because it converts more directly into capability (the FineWeb-Edu 8x multiplier [11]) and is far harder to substitute.

The thesis for an applied-AI lab is therefore to position as a data-acquisition and enrichment engine, not a model lab. Own or exclusively source the vertical corpus and its reward signal; rent the recipe. This is capital-efficient — it avoids the losing race to out-research a fast-following open community — and builds an asset whose value compounds as the public well runs dry. The defensibility test for any vertical-AI investment reduces to one question: does the company own data its competitors cannot get? Where the answer is yes, the moat survives every method cycle.

9. Limitations and Counter-Evidence

The claim is a trend, not a theorem, and has boundary conditions. First, it is strongest in data-rich verticals; Gu et al. explicitly condition the from-scratch advantage on abundant in-domain text [1]. In genuinely data-poor domains, synthetic generation or transfer may dominate, partially re-shifting value toward method — and synthetic-data methods are themselves advancing (a ~30% synthetic / 70% natural pretraining mixture has been shown to reach the same validation loss 5-10x faster), a live counter-current to pure data scarcity [18].

Second, our cross-vertical comparison holds method only approximately constant — Llemma and DeepSeekMath differ in both corpus and pipeline, and the two Llemma models differ in token budget (200B for 7B, 50B for 34B) [3][4]. We mitigate this with the FineWeb experiment, which does control architecture, scale, and method [11], but that experiment is in general web pretraining, not a high-stakes vertical, so external validity to e.g. clinical or legal corpora is an inference, not a measurement.

Third, several headline numbers carry caveats reported here in full. DeepSeekMath's 51.7% is no-tool greedy; its 60.9% requires 64-sample self-consistency [4]. Med-PaLM 2's 86.5% reflects an ensemble-refinement prompting strategy, and the improvement over the prior state of the art depends on the chosen baseline (the original Flan-PaLM 540B reports 67.6%; the separately fine-tuned Med-PaLM reports 67.2%) [2]. PubMedBERT's BLURB score is 81.35 under standard fine-tuning, rising to 82.91 with optimized fine-tuning [1] — a reminder that method still moves results at the margin even when corpus dominates.

Fourth, the licensing figures are reported or estimated terms, not audited financials, and a curated vertical corpus may price very differently from a social corpus [10][13]. Fifth, a genuine counter-thesis exists: if frontier gains came predominantly from architecture or test-time-compute scaling rather than data, the corpus moat would weaken — and the 2025-26 long-reasoning-RL wave makes this concrete (DeepSeek-V3.2 lifts an open model to GPT-5-class reasoning and 2025 IMO/IOI gold chiefly by scaling RL compute [16]). We note that even those regimes consume verifier and environment data as their binding input: in the controlled RLVE study, jointly training over 400 verifiable environments yields a +3.37% absolute reasoning gain while merely continuing the original RL run at 3x the compute yields only +0.49% [19] — i.e. environment/data scaling, not raw compute, drives the gains, which re-routes value back to whoever owns the verifiable corpus. We flag this as the central risk to the thesis and the variable to monitor.

10. Conclusion

Read individually, PubMedBERT, Med-PaLM 2, Llemma, DeepSeekMath, and SaulLM each tell a local story about a clever corpus. Read together, and stress-tested against the FineWeb experiment that isolates the corpus variable under full control, they reveal an invariant: across medicine, mathematics, finance, code, and law, domain capability tracks the domain corpus, and the methods that consume it commoditize within months [1][2][3][4][11][12]. Against a backdrop of a finite, near-exhausted public text stock and diminishing returns to repetition [5][7], exclusivity over high-signal vertical data — and over the reward signal that grades it [6] — is the asset that holds value through method churn.

For Perpetual Labs the directive is operational: concentrate capital and exclusivity on owning or exclusively sourcing the domain corpus, its extraction/quality pipeline, and its reward signal; adopt the modeling method as a commodity. The moat is the data, not the recipe.

INVESTMENT THESIS

Underwrite the data, not the recipe. The capability frontier in every vertical we examined (medicine, math, finance, code, law) tracks the in-domain corpus, and the FineWeb-Edu natural experiment proves it under full control: better data is an ~8x token (and compute) multiplier with architecture and method held fixed. Methods (GRPO, LoRA, process RM) go from paper to free open clone in months — GRPO's scaled-RL successor is already open, carrying DeepSeek-V3.2 to GPT-5-class reasoning and 2025 IMO/IOI gold [16] — while corpora, extraction pipelines, and reward-label sets (PRM800K, FinPile) do not, and the 2025-26 curation literature now replicates the FineWeb-Edu control (same loss asymptote on 5.17x less data [15]). With the ~300T-token public stock projected to be fully utilized between 2026 and 2032 and a real licensing market already pricing generic social data at $60-70M/yr (and up to $250M for premium archives), exclusive high-signal vertical data is the appreciating, non-depreciating layer of the stack. The one diligence question that predicts a durable moat: does the company own or exclusively source data — and the pipeline/reward signal that grades it — that competitors cannot obtain? Fund that; assume the model is a commodity.

References

[1] Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J., & Poon, H. (2021). Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing (PubMedBERT; BLURB benchmark). ACM Transactions on Computing for Healthcare. arXiv:2007.15779. BLURB macro avg 81.35 (82.91 with optimized FT).

[2] Singhal, K., Tu, T., Gottweis, J., et al. (2023). Towards Expert-Level Medical Question Answering with Large Language Models (Med-PaLM 2). arXiv:2305.09617. MedQA up to 86.5%; original Flan-PaLM 540B 67.6% (arXiv:2212.13138).

[3] Azerbayev, Z., Schoelkopf, H., Paster, K., et al. (2023). Llemma: An Open Language Model for Mathematics. arXiv:2310.10631. MATH 4-shot greedy: Llemma-34B 25.0% vs Code Llama-34B 12.2%; Llemma-7B 18.0% vs 4.5%. Proof-Pile-2 = 55B tokens.

[4] Shao, Z., Wang, P., Zhu, Q., et al. (2024). DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models (introduces GRPO). arXiv:2402.03300. MATH 51.7% (no tools); 60.9% SC@64; 120B mined math tokens.

[5] Villalobos, P., Ho, A., Sevilla, J., Besiroglu, T., Heim, L., & Hobbhahn, M. (2024). Will We Run Out of Data? Limits of LLM Scaling Based on Human-Generated Data. ICML 2024 (Position). arXiv:2211.04325 / Epoch AI. ~300T-token effective stock (90% CI 100T-1000T); full utilization 2026-2032.

[6] Lightman, H., Kosaraju, V., Burda, Y., et al. (2023). Let's Verify Step by Step (process supervision; PRM800K, 800K step-level labels). arXiv:2305.20050. 78.2% of MATH subset via best-of-N (vs 72.4% outcome supervision).

[7] Muennighoff, N., Rush, A.M., Barak, B., et al. (2023). Scaling Data-Constrained Language Models. NeurIPS 2023. arXiv:2305.16264. Up to ~4 epochs negligible loss change; with further repetition the value of added compute decays to zero.

[8] Wu, S., Irsoy, O., Lu, S., et al. (2023). BloombergGPT: A Large Language Model for Finance. arXiv:2303.17564. FinPile 363B proprietary financial tokens; ~708B total corpus (+345B general); 50B params.

[9] Hu, E.J., Shen, Y., Wallis, P., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.

[10] Columbia Journalism Review (2025), 'Reddit Is Winning the AI Game'; Variety/WSJ (2024) on OpenAI-News Corp (up to $250M/5yr); Reddit-Google ~$60M/yr, Reddit-OpenAI ~$70M/yr, ~10% of Reddit revenue (~$130M).

[11] Penedo, G., Kydlicek, H., Allal, L.B., et al. (2024). The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale (FineWeb 15T; FineWeb-Edu 1.3T). arXiv:2406.17557. MMLU 33%->37%, ARC 46%->57%; 33.6% MMLU at 38B vs ~300B tokens for Matrix (Fig 11).

[12] Colombo, P., Pires, T.P., Boudiaf, M., et al. (2024). SaulLM-7B: A Pioneering Large Language Model for Law. arXiv:2403.03883. 30B+-token English legal corpus over Mistral-7B; SOTA on legal-document tasks among open models. See also SaulLM-54B/141B, arXiv:2407.19584.

[13] Fortune Business Insights (2025), AI Training Dataset Market ~$3.59B (2025), ~$4.44B (2026), CAGR ~22.9% to 2034.

[14] Yang, A., Zhang, B., Hui, B., et al. (2024). Qwen2.5-Math Technical Report: Toward Mathematical Expert Model via Self-Improvement. arXiv:2409.12122. Qwen Math Corpus expanded from 700B (v1) to over 1T tokens (v2).

[15] Kim, K., Kotha, S., Liang, P., & Hashimoto, T. (2025). Pre-training under infinite compute. arXiv:2509.14786. A regularized + ensembled recipe reaches the same validation-loss asymptote using 5.17x less data than the baseline (optimal weight decay ~30x larger),

with model and optimizer fixed — a controlled corroboration of FineWeb-Edu's data-quality control.

[16] DeepSeek-AI (2025). DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models. arXiv:2512.02556. Scaling RL post-training (the GRPO-family recipe) lifts an open model to GPT-5-class reasoning; the DeepSeek-V3.2-Speciale variant surpasses GPT-5, reaches parity with Gemini-3.0-Pro, and wins gold-medal performance at the 2025 IMO and IOI — the realized open successor to GRPO.

[17] Mizrahi, D., Larsen, A.B.L., Allardice, J., Petryk, S., et al. (Apple) (2025). Language Models Improve When Pretraining Data Matches Target Tasks (BETR). arXiv:2507.12466. Selecting pretraining documents by embedding-similarity to target-task data improves nearly every downstream benchmark across scales — confirming that matching the corpus to the target, not the method, sets capability.

[18] Mixture study on synthetic + natural pretraining data (>1000 LLMs) (2025). arXiv:2510.01631. A ~1/3 rephrased-synthetic, 2/3 natural-web mixture reaches the same validation loss 5-10x faster at larger data budgets — quantifying the synthetic-data counter-current to pure data scarcity.

[19] Zeng, Z., Ivison, H., Wang, Y., Yuan, L., et al. (2025). RLVE: Scaling Up Reinforcement Learning for Language Models with Adaptive Verifiable Environments. arXiv:2511.07317. Jointly training over 400 verifiable environments yields +3.37% absolute reasoning gain vs +0.49% from merely continuing the original RL run at 3x compute — environment/data scaling, not raw compute, drives RL gains.